

TEAU 1340 – Manual Drive Trains and Axles (3 Credits)

Course Description

Manual Drive Trains is an in-depth clinical based course that conforms to the ASE/NATEF standards. Throughout the course, you will concentrate on topics such as safety concerns when working on drive train components, power flow through a gear set, clutch diagnosis and operation, manual gearbox diagnosis, removal and rebuilding procedures for manual transmissions, 4-wheel drive systems. When you have completed this course, you will be prepared to take the certification exam in ASE (Automotive Service Excellence) Automotive Manual Drive Train Systems.

Course Objectives

- Demonstrate a working knowledge of manual drivetrains and axle systems.
- Diagnose, remove, repair, and replace manual transmission and components.
- Remove driveshafts, replace universal joints and repair four-wheel drive systems.
- Diagnose and repair electronically controlled transmission systems.

Course Outline

- Manual Drivetrain and Axle Theories
- Clutches
- Manual Transmissions, Transaxles, and Drive Axles
- Drive Shafts, Universal Joints, and Drive Axles
- Four-Wheel Drive Systems
- Advanced Four-Wheel Drive Systems
- Drivetrain Electrical and Electronic Systems
- Electronically-Controlled and Automatic Trans

Textbook & Reading Materials

Today's Technician: Manual Transmissions and Transaxles Classroom Manual and Shop Manual, 7th ed. (ISBN: 9781337795456)

Assignments & Assessments

Orientation	CM CH 2 Labeling Activity 2-1: Gears and Torque Multiplication ID #1
Orientation Acknowledgement	CM CH 2 Labeling Activity 2-2: Gears and Torque Multiplication ID #2
Remind Txt Group	CM CH 2 Labeling Activity 2-3: Transmission Gear Ratios ID
STECH Auto Student Information Sheet	Setting up a Dial Indicator Simulation
STECH COVID-19 Policies and Procedures Agreement Review	CM CH 2 Fill-in-the-Blank Questions: Drivetrain Theory
STECH Auto Right -to-Know agreement	CM CH 2 Multiple-Choice Questions: Drivetrain Theory
SWAM 2820 Manual Drive Trains and Axles Course Syllabus	SM CH 2 Reading: Special Tools and Procedures
Review	SM CH 2 Labeling Activity 2-4: Micrometer Parts ID
SWAM 2820 Manual Drive Trains and Axles Lab Assignment	SM CH 2 Labeling Activity 2-5: Dial Indicator Parts ID
Checklist Review	SM CH 2 Photo Sequence 2 - Setting Up a Dial Indicator
SWAM 2820 Manual Drive Trains and Axles Competency Profile and Task List Review	SM Chapter 2 ASE-Style Review Questions: Special Tools and Procedures
Cleaning Expectations	MDT LAB 1 Identify and interpret drive train concerns, determine necessary action
Southwest Technical College Automotive Video Playlist	MDT LAB 2 Identify and interpret drive train concerns, determine necessary action
Student Tool and Equipment Use Waiver	MDT LAB 3 Research applicable vehicle and service information, fluid type, vehicle service history, service precautions, and technical service bulletins
Automotive Student OE Instructions	MDT LAB 4 Check fluid condition; check for leaks; determine necessary action
Cell Phone	MDT LAB 5 Check fluid condition; check for leaks; determine necessary action
Instructions	MDT LAB 6 Drain and refill manual transmission/ transaxle and final drive unit
Digital Lab Explanation	MDT LAB 7 Complete work order to include customer information, vehicle identifying
Module Breakdown	Checkpoint Meeting Module 1
Module 1 Labs	End Module 1
Module 2 Labs	CM CH 3 Reading: Clutches
Module 3 Labs	How a Clutch Operates Simulation
Module 4 Labs	How a Dual Clutch Transmission Operates Simulation
Module 5 Labs	CM CH 3 Labeling Activity 3-1: Clutch Assembly Parts ID
Module 6 Labs	CM CH 3 Labeling Activity 3-2: Hydraulic Clutch Linkage ID
Module 7 Labs	CM CH 3 Video 3-1 Questions: Single Clutch
Module 8 Labs	CM CH 3 Video 3-2 Questions: Throw Out Bearings
Cleaning Lab Module	CM CH 3 Video 3-3 Questions: Clutch Linkages
Fillable Repair Order Document	CM Chapter 3 Fill-in-the-Blank Questions: Clutches
Cleaning Lab 1	CM Chapter 3 Multiple-Choice Questions: Clutches
Cleaning Lab 2	SM CH 3 Reading: Clutch Service
Cleaning Lab 3	SM CH 3 Photo Sequence 3 - Bleeding a Hydraulic Clutch
Cleaning Lab 4	
Cleaning Lab 5	
CM CH 1 Reading: Intro to Manual Drive Trains and Axles	
CM CH 1 Labeling Activity 1-1: Clutch Parts ID	
CM CH 1 Labeling Activity 1-2: Axle and Bearing ID	
CM CH 1 Video 1-1 Questions: Types of Gears	
CM Chapter 1 Fill-in-the-Blank Questions: Intro to Manual Drive Trains	
CM Chapter 1 Multiple-Choice Questions: Intro to Manual Drive Trains	
CM CH 2 Reading: Drivetrain Theory	

SM CH 3 Photo Sequence 4 - Installing and Aligning a Clutch

Disc

SM CH 3 Video 3-4 and Questions: Check Clutch

SM CH 3 ASE-Style Review Questions: Clutch Service

SM CH 3 ASE Challenge Questions: Clutch Service

MDT LAB 8 Diagnose clutch noise, binding, slippage, pulsation, and chatter; determine necessary action

MDT LAB 9 Inspect clutch pedal linkage, cables, automatic adjuster mechanisms, brackets, bushings, pivots, and springs; perform necessary action

MDT LAB 10 Inspect clutch pedal linkage, cables, automatic

adjuster mechanisms, brackets, bushings, pivots, and springs; perform necessary action

MDT LAB 11 Inspect and replace clutch pressure plate assembly, clutch disc, release (throw-out) bearing and linkage, and pilot bearing/bushing (as applicable)

MDT LAB 12 Inspect and replace clutch pressure plate assembly, clutch disc, release (throw-out) bearing and linkage, and pilot bearing/bushing (as applicable)

MDT LAB 13 Check and adjust clutch master cylinder fluid level; check for leaks

MDT LAB 14 Inspect flywheel and ring gear for wear and cracks; determine necessary action

MDT LAB 15 Check and adjust clutch master cylinder fluid level; check for leaks

MDT LAB 16 Inspect flywheel and ring gear for wear and cracks; determine necessary action

MDT LAB 17 Inspect flywheel and ring gear for wear and cracks; determine necessary action

Checkpoint Meeting Module 2

End Module 2

CM CH 4 Reading: Manual Transmissions and Transaxles

CM CH 4 Labeling Activity 4-1: Synchronizer Parts Identification

CM CH 4 Labeling Activity 4-2: Transmission Parts Identification

Video 4-1 and Questions: Synchronizer Operation #1

Video 4-2 and Questions: Synchronizer Operation #2

Video 4-3 and Questions: Transmission Power Flow

Video 4-4 and Questions: Transaxle Power Flow

Gear Ratios Explained Simulation

CM CH 4 Fill-in-the-Blank Questions: Manual Transmissions

and Transaxles

CM CH 4 Multiple-Choice Questions: Manual Transmissions

and Transaxles

SM CH 4 Reading: Manual Transmissions/ Transaxles Servicing

SM CH 4 Labeling Activity 4-3: Synchronizer Parts Identification

SM CH 4 Video 4-5 and Questions: Manual Transmissions

SM CH 4 ASE-Style Review Questions: Transmission & Transaxle Service

SM CH 4 ASE Challenge Questions: Transmission & Transaxle Service

MDT LAB 18 Inspect, adjust, and install shift linkages, brackets, bushings, cables, pivots, and levers

MDT LAB 19 Inspect, adjust, and install shift linkages, brackets, bushings, cables, pivots, and levers

MDT LAB 20 Diagnose noise concerns through the application of transmission / transaxle power-flow principles

MDT LAB 21 Diagnose hard shifting and jumping out of gear

concerns; determine necessary action

MDT LAB 22 Diagnose transaxle final drive assembly noise

and vibration concerns; determine necessary action

Checkpoint Meeting Module 3

End Module 3

CM CH 5 Reading: Front Drive Axles

CM CH 5 Labeling Activity 5-1: CV Axle Parts Identification

CM CH 5 Labeling Activity 5-2: FWD Wheel Bearing & Hub Parts Identification

CM CH 5 Video 5-1 and Questions: Front Wheel Drive Axles

CM CH 5 Video 5-2 and Questions: CV Joints

How a Constant Velocity Joint-Axle Works on a FWD Simulation

CM CH 5 Fill-in-the-Blank Questions: Front Drive Axles

CM CH 5 Multiple-Choice Questions: Front Drive Axles

SM CH 5 Reading: Front Drive Axle Service

SM CH 5 Labeling Activity 5-4: Outer CV Joint Parts Identification

SM CH 5 Photo Sequence 9 - Removing and Installing Drive

Axles

SM CH 5 ASE-Style Review Questions: Front Drive Axle Service

SM CH 5 ASE Challenge Questions: Front Drive Axle Service

CM CH 6 Reading: Drive Shaft & Universal Joints

CM CH 6 Labeling Activity 6-1: RWD Drivetrain Parts Identification

CM CH 6 Labeling Activity 6-2: RWD Coil Spring Suspension

Identification

CM CH 6 Fill-in-the-Blank Questions: Drive Shaft & Universal

Joints

CM CH 6 Multiple-Choice Questions: Drive Shaft & Universal

Joints

SM CH 6 Reading: Servicing Drive Shafts & Universal Joints

SM CH 6 Photo Sequence 13 - U-Joint Service with Special Tool

SM CH 6 ASE Challenge Questions: Servicing Drive Shafts & Universal Joints

Chapter 7 Power Point Overview: Differentials & Drive Axles;

Differentials & Drive Axles Service

CM CH 7 Reading: Differentials & Drive Axles

How a Rear Differential Works Simulation

CM CH 7 Labeling Activity 7-1: Planetary Gears Identification

CM CH 7 Labeling Activity 7-2: Differential Parts Identification

CM CH 7 Video 7-1 and Questions: Differential and Drive Axles

CM CH 7 Video 7-2 and Questions: Differential Components

CM CH 7 Video 7-3 and Questions: Limited Slip Differentials

CM CH 7 Video 7-4 and Questions: Drive Axles and Bearings

CM CH 7 Fill-in-the-Blank Questions: Differentials & Drive Axles

CM CH 7 Multiple-Choice Questions: Differentials & Drive Axles

SM CH 7 Reading: Differential Diagnosis and Service

SM CH 7 Labeling Activity 7-3: Gear Contact Patterns

SM CH 7 Photo Sequence 20 The Assembly of an Integral carrier-Type Final Drive Unit

SM CH 7 Photo Sequence 21 Measuring and Adjusting Backlash and Side Bearing Preload

Measuring Backlash on a Differential Assembly Simulation

SM CH 7 ASE-Style Review Questions: Differential Diagnosis

and Service

SM CH 7 ASE Challenge Questions: Differential Diagnosis and Service

MDT LAB 23 Diagnose constant -velocity (CV) joint noise and

vibration concerns;

determine necessary action

MDT LAB 24 Diagnose constant -velocity (CV) joint noise and

vibration concerns;

determine necessary action

MDT LAB 25 Diagnose universal joint noise and vibration concerns: determine necessary action

MDT LAB 26 Inspect, remove, and replace front wheel drive

(FWD) bearings, hubs, and

seals

MDT LAB 27 Inspect, service, and replace shafts, yokes, boots, and universal/CV joints

MDT LAB 28 Check shaft balance and phasing; measure shaft runout ; measure and adjust

driveline angles

MDT LAB 29 Clean and inspect differential housing; check for leaks; inspect housing vent

MDT LAB 30 Check and adjust differential housing fluid level

MDT LAB 31 Drain and refill differential housing

MDT LAB 32 Diagnose noise and vibration concerns; determine necessary action

MDT LAB 33 Inspect and replace companion flange and pinion seal; measure companion flange runout

MDT LAB 34 Diagnose noise, slippage, and chatter concerns;

determine necessary action

MDT LAB 35 Inspect and replace drive axle wheel studs

MDT LAB 36 Remove and replace drive axle shafts

MDT LAB 37 Inspect and replace drive axle shaft seals, bearings, and retainers

MDT LAB 38 Measure drive axle flange runout and shaft end

play; determine necessary action

MDT LAB 39 Diagnose drive axle shafts, bearings, and seals

for noise, vibration, and fluid leakage concerns, determine necessary action

Checkpoint Meeting Module 4

End Module 4

CM CH 8 Reading: Four-Wheel-Drive Systems

CM CH 8 Labeling Activity 8-1: 4WD Drivetrain Component Identification

CM CH 8 Labeling Activity 8-2: Transfer Case Component Identification

CM CH 8 Video 8-1 Questions: Four-Wheel-Drive Systems

CM CH 8 Video 8-2 Questions: 4WD Systems Components

CM CH 8 Video 8-3 Questions: Transfer Case Designs
 CM CH 8 Fill-in-the-Blank Questions: Four-Wheel-Drive Systems
 CM CH 8 Multiple-Choice Questions: Four-Wheel-Drive Systems
 SM CH 8 Reading: Servicing Four-Wheel-Drive Systems
 SM CH 8 Labeling Activity 8-3: Transfer Case Parts Identification
 SM CH 8 Labeling Activity 8-4: 4WD Front Hub Parts Identification
 SM CH 8 Photo Sequence 24 - Assembling a Warner 13-50 Transfer Case
 SM CH 8 ASE-Style Review Questions: Servicing Four-Wheel Drive Systems
 SM CH 8 ASE Challenge Questions: Servicing Four-Wheel Drive Systems
 MDT LAB 40 Inspect, adjust, and repair shifting controls (mechanical, electrical, and vacuum), bushings, mounts, levers, and brackets
 MDT LAB 41 Inspect front-wheel bearings and locking hubs;
 perform necessary action(s)
 MDT LAB 43 Identify concerns related to variation in tire circumference and/or final drive ratios
 MDT LAB 44 Diagnose noise, vibration, and unusual steering concern; determine necessary action
 MDT LAB 45 Diagnose, test, adjust, and replace electrical/electronic components of four-wheel drive systems
 Checkpoint Meeting Module 5
 End Module 5
 CM CH 9 Reading: Advanced 4WD Systems
 CM CH 9 Labeling Activity 9-1: Viscous Clutch Parts Identification
 CM CH 9 Labeling Activity 9-2: Hydraulic Clutch Control Circuit
 CM CH 9 Fill-in-the-Blank Questions: Advanced 4WD Systems
 CM CH 9 Multiple-Choice Questions: Advanced 4WD Systems
 SM CH 9 Reading: Diagnosing 4WD and AWD Control Systems
 SM CH 9 Labeling Activity 9-3: Differential Parts Identification
 SM CH 9 Photo Sequence 25 - Diagnosis with a Scan Tool
 SM CH 9 ASE-Style Review Questions: Diagnosing 4WD and AWD Control Systems

SM CH 9 ASE Challenge Questions: Diagnosing 4WD and AWD Control Systems
 MDT LAB 46 Describe the operation and service of a system that uses a dual mass flywheel
 MDT LAB 47 Measure rotating torque on a limited slip differential; determine needed
 MDT LAB 48 Disassemble, inspect, service, and reassemble a four wheel drive transfer case and components
 MDT LAB 49 Disassemble, inspect, service, and reassemble a limited-slip rear wheel drive rear axle / differential assembly
 MDT LAB 50 Ensure vehicle is prepared to return to customer per school/company policy
 MDT LAB 51 Disassemble, inspect, service, and reassemble a rear wheel drive manual transmission
 MDT LAB 52 Disassemble, inspect, service, and reassemble a rear wheel drive manual transmission
 Checkpoint Meeting Module 6
 End Module 6
 CM CH 10 Reading: Drivetrain Electrical and Electronic Systems
 Using the Voltmeter Simulation
 Voltage Drop Testing Simulation
 Using an Ohmmeter Simulation
 Reading Wiring Diagrams Simulation
 CM CH 10 Labeling Activity 10-1: Simple Circuit Diagram
 CM CH 10 Labeling Activity 10-2: Clutch Switch Parts Identification
 CM CH 10 Fill-in-the-Blank Questions: Drivetrain Electrical and Electronic Systems
 CM CH 10 Multiple-Choice Questions: Drivetrain Electrical and Electronic Systems
 SM CH 10 Reading: Servicing Drivetrain Electrical Systems
 SM CH 10 Photo Sequence 26 - Voltage Drop Test to Locate High Circuit Resistance
 SM CH 10 ASE-Style Review Questions: Servicing Drivetrain Electrical Systems
 SM CH 10 ASE Challenge Questions: Servicing Drivetrain Electrical Systems
 MDT LAB 53 Disassemble, inspect, service, and reassemble a rear wheel drive manual transmission

MDT LAB 54 Disassemble, inspect, service, and reassemble a rear wheel drive manual transmission

MDT LAB 55 Disassemble, inspect, service, and reassemble a four wheel drive transfer case and components

MDT LAB 56 Disassemble, inspect, service, and reassemble a rear wheel drive rear axle/ differential assembly

Checkpoint Meeting Module 7

End Module 7

CM CH 11 Reading: Electronically Controlled and Automated Transmissions

CM CH 11 Labeling Activity 11-1: Hybrid Vehicle Component Identification

CM CH 11 Labeling Activity 11-2,: Advanced Transfer Case Parts Identification

CM CH 11 Fill-in-the-Blank Questions: Electronically Controlled and Automated Transmissions

CM CH 11 Multiple-Choice Questions: Electronically Controlled and Automated Transmissions

SM CH 11 Reading: Diagnosing Advanced Electronic Systems

SM CH 11 Labeling Activity 11-3:

SM CH 11 Photo Sequence 27 - Flashing a PCM or BCM

SM CH 11 ASE-Style Review Questions: Diagnosing Advanced Electronic Systems

SM CH 11 ASE Challenge Questions: Diagnosing Advanced Electronic Systems

MDT LAB 57 Describe the operational characteristics of an electronically-controlled manual transmission/transaxle

MDT LAB 58 Disassemble, inspect, service, and reassemble a rear wheel drive rear axle/ differential assembly

MDT LAB 59 Disassemble, inspect, service, and reassemble the front axle/ differential assembly of a four wheel drive vehicle

MDT LAB 60 Disassemble, inspect, service, and reassemble the front axle/ differential assembly of a four wheel drive vehicle

Checkpoint Meeting Module 8

End Module 8

End of Course Survey

Student Feedback

Manual Drive Trains and Axles Competency Profile (2021)

Manual Transmissions & Transaxles Review

Manual Transmissions & Transaxles Final Exam

Subject to change. Please consult your Canvas course for the most current instructions and updates.

Classroom Hours

Mo, Tu, W, Th

8:00 AM - 12:00 AM

1:00 PM - 5:00 PM

Friday

8:00 AM - 12:00 AM

For a full list of course hours visit: [Course Schedule](#)

Instructor Contact Information

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Office Hours: By appointment

Email is the preferred method of communication; you will receive a response within 24 hours during regular business hours.

Canvas Information

Canvas is where course content, grades, and communication will reside for this course.

- Canvas URL: stech.instructure.com
 - For Canvas passwords or any other computer-related technical support contact Student Services.
 - For Regular Hours and Weekdays call (435) 586 - 2899
 - For After Hours and Weekends call (435) 865 - 3929 (Leave a message if no response)
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Policies

Course Grading

Course Grading: Students must achieve 80% (B-) or higher to pass graded work. Incomplete assignments must be redone to meet the required standards. Guidelines, rules, and expectations for completing assignments are provided in each course.

High School Power School Grades: Quarter student grades will be determined by student progress percentage. Faculty will use the higher percentage of either 1) quarter progress, or 2) cumulative progress for the current training plan year.

Grade Scale: The following grading scale will be used to determine a letter grade.

A : 94 - 100%	A- : 90 - 93%	B+ : 87 - 89%	B : 83 - 86%
B- : 80 - 82%	C+ : 77 - 79%	C : 73 - 76%	C- : 70 - 72%
D+ : 67 - 69%	D : 63 - 66%	D- : 60 - 62%	F : 0 - 59%

Course Policies

Class attendance is required during your scheduled time. Cell phones for many have become a distraction. When you are in class or lab we encourage you to keep your cell phones put away in a secure location. If you use ear buds we ask that you only use one so you can still hear the things going on around you. If you are using your phone for things other than school related items, instructors will ask you to put them away. Please follow the direction of your instructors. Those who have been asked to refrain from using your cell phone and fail to do so will be asked to meet with the Director of Transportation and student services will be notified. The program is designed to provide the student with as much hands-on work as possible. In the automotive industry you may be required to lift heavy objects and stand for hours at a time to complete work required. Technicians deal with chemicals and materials which require caution, these will be identified in the Right to Know Agreement provided to you. You will also be required to use computers to track and complete work.

Additional Information

InformaCast Statement: Southwest Tech uses InformaCast to ensure the safety and well-being of our students. In times of emergency, such as weather closures and delays, this app allows us to promptly deliver notifications directly to your mobile devices. To stay informed and receive real-time updates, we encourage all students to sign up for notifications. Your safety is our priority, and staying connected ensures a swift response to any unforeseen circumstances. More information and directions for signing up are available at: <https://stech.edu/emergency-notifications/>

Internet Acceptable Use Policy: The student is expected to review and follow the Southwest Technical College Internet Safety Policy at: <https://stech.edu/students/policies/>

Student Code of Conduct Policy: The student is expected to review and follow the Southwest Technical College Student Code of Conduct Policy at: <https://stech.edu/students/policies/>

Accommodations: Students with medical, psychological, learning, or other disabilities desiring accommodations or services under ADA, must contact the Student Services Office. Student Services determines eligibility for and authorizes the provision of these accommodations and services. Students must voluntarily disclose that they have a disability, request an accommodation, and provide documentation of their disability. Students with disabilities may apply for accommodations, based on an eligible disability, through the Student Services office located at 757 W. 800 S., Cedar City, UT 84720, and by phone at (435) 586-2899. No diagnostic services are currently available through Southwest Technical College.

Safety and Building Maintenance: The College has developed and follows a variety of plans to ensure the safe and effective operation of its facilities and programs. The following plans are available online: 1) Facilities Operations and Maintenance Plan; 2) Technical Infrastructure Plan; and 3) Health and Safety Plan.

Withdrawals and Refunds: Please refer to the Southwest Technical College Refund Policy at: <https://stech.edu/students/policies/>

Any high school or adult student, who declares a technical training objective is eligible for admission at Southwest Technical College (Southwest Tech). Program-specific admissions requirements may exist and will be listed on the Southwest Tech website. A high school diploma or equivalent is not required for admission but is mandatory for students seeking Title IV Federal Financial Aid.

Non-Discriminatory Policy: Southwest Technical College affirms its commitment to promote the goals of fairness and equity in all aspects of the educational enterprise, and bases its policies on the idea of global human dignity.

Southwest Tech is committed to a policy of nondiscrimination. No otherwise qualified person may be excluded from participation in or be subjected to discrimination in any course, program or activity because of race, age, color, religion, sex, pregnancy, national origin or disability. Southwest Technical College does not discriminate on the basis of sex in the education programs or activities that it operates, as required by Title IX and 34 CFR part 106. The requirement not to discriminate in education programs or activities extends to admission and employment. Inquiries about Title IX and its regulations to STECH may be referred to the Title IX Coordinator, to the Department of Education, and/or to the Office for Civil rights.

If you believe you have experienced discrimination or harassment on our campus, please contact the Title IX Coordinator, Cory Estes: cestes@stech.edu, (435) 865-3938.

For special accommodations, please contact the ADA Coordinator, Cyndie Tracy: ctracy@stech.edu, (435) 865-3944.

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